ABSTRACT

This project presents a innovative solution for regulating fan speed based on ambient temperature, utilizing an Arduino Uno board as the central processing unit. By integrating a temperature sensor to detect changes in temperature, the system automatically adjusts the fan speed to maintain a comfortable environment. The project aims to optimize energy consumption, reduce noise pollution, and enhance user experience. Through a user-friendly interface, the system displays real-time temperature readings and fan speed status. The temperature-based fan speed control system offers a cost-effective, efficient, and eco-friendly solution for various applications, including residential, commercial, and industrial settings. By harnessing the power of automation and sensor technology, this project demonstrates a cutting-edge approach to sustainable and intelligent environmental control.

CHAPTER 1:

1.1 INTRODUCTION

The pursuit of creating a more sustainable and comfortable living environment, the integration of intelligent control systems has become a crucial aspect of modern technology. One area of focus is the development of efficient fan speed control systems, which can significantly impact energy consumption and noise levels. Traditional fan control methods often rely on simple on/off switches or basic thermostatic controls, which can lead to energy wastage and reduced comfort. To address these limitations, this project introduces a novel temperature-based fan speed control system, designed to optimize fan performance and energy efficiency. By harnessing the power of advanced sensor technologies and microcontroller systems, this innovative solution can dynamically adjust fan speed in response to changing ambient temperatures, ensuring a comfortable environment while minimizing energy consumption and noise pollution. With its potential applications spanning across residential, commercial, and industrial settings, this project aims to contribute to the growing field of sustainable and intelligent environmental control systems.

1.2 OBJECTIVES

- Designing a temperature-based fan speed control system to regulate fan speed.
- Optimize fan performance to maintain a comfortable environment.
- Fan speed adjustment using temperature sensors.
- Reduce energy consumption by minimizing fan speed.
- Improve user experience with a quiet and comfortable environment.
- Develop a cost-effective solution for temperature control.
- Create a user-friendly interface for temperature monitoring.
- Ensure system reliability and durability for long-term use.
- Implement a scalable solution for various applications.
- Enhance system efficiency with real-time temperature feedback.

1.3 PROBLEM STATEMENT

Temperature based fan speed controller using Arduino: Many existing systems for temperature monitoring and controlling generally uses micro-controller ATMEL 89C51 (µc 8051). Due to using micro controller 8051 the process of making whole device becomes not only very complex but also difficult and tedious. For operation it requires A-D converter, external clock, microcontroller development board. Consequently, the problems are as follows:

a) It takes comparatively more time to process.

b) It requires additional devices for operation.

c) It requires external clock.

d) Programming for microcontroller 8051 is difficult.

e) For programming it requires development system.

CHAPTER 2

LITERATURE SURVEY

1. **K Sai Kumar, Sirolla Rajasekhar, Shaik Haseeb Tanveer,S Venkata Prasad Reddy, Valthati Allababu, Automatic temperature based fan controller,2023**
   - In this work they have designed the Temperature based Speed controlled fan. The surrounding temperature is sensed by LM35 sensor the sensed signal is sent as voltage pulses to microcontroller ATMEGA 328P and converts into an electrical signal which is applied to micro controller.
   - The micro controller on arduino drives the motor driver to control the fan. ESP8266 Wi-Fi module is interfaced with arduino by using UBIDOTS free cloud storage website it is possible to switch the fan using mobile phone or laptop.
   - Temperature variations, after sometimes its efficiency decreases.
2. **Shivshankar Adsule, Shivani Mohite, Rahul Patil, Namrata R. Dhawas, Automatic Temperature Based Fan Speed Controller Using Arduino,2020**
   - Temperature Based Fan Speed Control & Monitoring with Arduino and LM35 Temperature Sensor.
   - The microcontroller controls the speed of an electric fan according to the requirement & allows dynamic and faster control and the LCD makes the system user-friendly.
   - Sensed temperature in Celsius Scale and fan speed in percentage are simultaneously displayed on the LCD panel.
   - Expensive, complex in design and not flexible.
3. **Srinivas P, Kavinkumar B, Arun Venkat A, Dr.R.Senthil Kumar, Temperature based Fan Speed Controller,2020**
   - In the proposed system itself said that it is designed to detect the temperature of the room and send that information to the Arduino UNO board. Then the Arduino UNO board carries out the contrast of current temperature and set temperature based on the inbuilt program of the Arduino that feed through us.
   - The output obtained from the operation is given through the o/p port of an Arduino UNO board to the LCD display that connected with the board.
   - Less efficiency, expensive and complex in design.
4. **B.L.Tanmai, K.Rishitha,M.Padma, SK.Khamurunnisa, Temperature controlled fan using ,2020**
   - This paper presents the design and simulation of the fan speed control system using PWM technique based on the room temperature. A temperature sensor has been used to measure the temperature of the room and the speed of the fan is varied according to the room temperature using PWM technique
   - No monitoring of speed on LCD display.

CHAPTER 3

3.1 METHODOLOGY

Advanced microcontroller called Arduino (ATmega8) has in-built with many components like analog to digital converter, clock of 16 MHz, shift registers. In this system we uses temperature sensor LM35, to use to detect temperature into appropriate voltage. This voltage is given to Arduino. According to program it processes the analog signal into digital and forms a particular voltage level for a particular temperature.16x2 LCD is used to display the output i.e. surrounding temperature of LM35 in both degree centigrade and Fahrenheit units. At the same time, it also sends the data to Relay, if the temperature becomes maximum from set point relay becomes activate and it switches on the cooling device like fan. In this manner it monitors and controls the temperature. Temperature is the most often-measured environmental quantity. This might be expected since most physical, electronic, chemical, mechanical and biological systems are affected by temperature. Some processes work well only within a narrow range of temperatures. Certain chemical reactions, biological processes, and even electronic circuits perform best within limited temperature ranges. When these processes need to be optimized, control systems that keep temperature within specified limits or constant are often used. The field of process control has grown rapidly since its inception in the 1959s. It has become one of the core areas of chemical engineering. One of the most important process variables to be controlled is temperature of liquid in many chemical engineering Plants. In brief, how to control the temperature of a water bath is one of the industrial problems. Hence suitable temperature controller for controlling temperature of a water bath is very common requirement of the industries. Temperature rise due to resultant increase in molecular activity of substances on application of heat, which also connotes an increase in internal energy of the material. Heat exchange, heat balances, safe temperature limits of various equipment’s and multitude of other problems involving temperature are important in all phases of power generation and process industries. The means of indicating, recording and controlling temperatures are important in design, fabrication, operation, and testing power generating and utilizing equipment and associated apparatus. Correct measurement of temperature is of paramount importance to power and process engineers and great strides have been made in this branch. The ultimate aim is to measure temperatures correctly and reliably, at the same time keeping the cost of instruments down.

3.2 PROPOSED BLOCK DIAGRAM

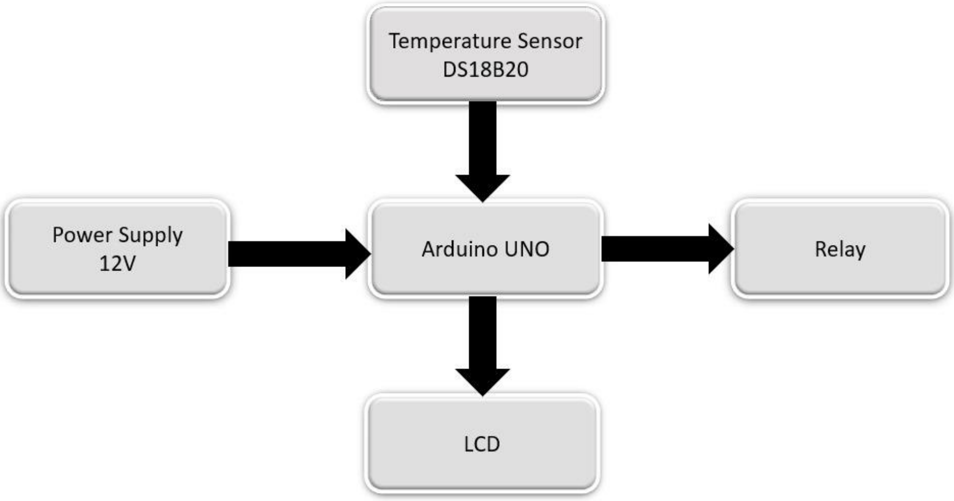

Fig.3.1: Block diagram

The fig.3.1 depicts the block diagram of the temperature based fan speed controlling using arduino.The function of each equipment depends on each other. The function of temperature sensor is to sense the temperature from the environment and give the analog input to the microcontroller at the Port C where ADC pin converts the analog signal into digital signals. Microcontroller has 28 pins. Some of the pins are known as VCC, GND, AVCC, etc. AVCC is used for ADC. There are three ports in the ATmega8 microcontroller which are Port B, Port C and Port D. each port can be used as input or output ports. In the project we used Port B as an output and output from the port is given to the motor driver. Pin PB1 is connected to the input 1 and pin PB2 is connected to the input 2 of the driver IC. The output of the motor driver IC is connected to the DC motor. DC motor runs when input 1 and input 2 is either 01(low high) or 10(high low).

3.3 CIRCUIT DIAGRAM AND WORKING

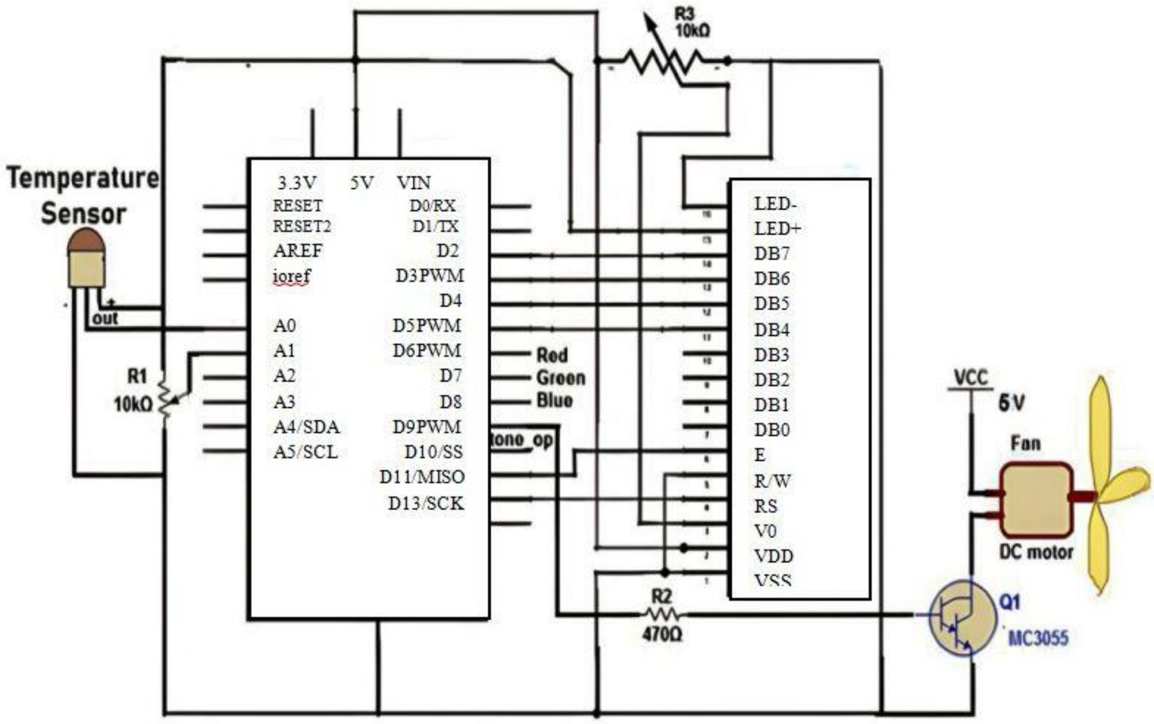

Fig.3.2: Circuit diagram

WORKING

Temperature Sensing: The DS18B20 sensor continuously monitors the ambient temperature and sends this data to the Arduino via its single wire communication protocol.

Data Processing: The Arduino reads the temperature data from the DS18B20 sensor. It then processes this data to determine if the temperature exceeds a predefined threshold.

Fan Control: Based on the temperature reading, the Arduino decides whether to turn the fan on or off:

- If the temperature is below the threshold, the Arduino keeps the fan off.
- If the temperature exceeds the threshold, the Arduino activates the relay, which in turn powers the DC fan to cool down the environment.

Display: The current temperature and the fan status (on/off) are displayed on the LCD.

CHAPTER 4

4.1 COMPONENT REQUIREMENTS AND DESCRIPTION

1. Arduino UNO
2. Power supply 12V
3. LCD
4. Jumper wires
5. DS18B20 single wire temperature sensor
6. Relay
7. DC Fan

DESCRIPTION:

1)Arduino Uno

Arduino is an open-source hardware and software company, project and user community that designs and manufactures single-board microcontrollers and microcontroller kits for building digital devices and interactive objects that can sense and control both physically and digitally. Its products are licensed under the GNU Lesser General Public License (LGPL) or the GNU General Public License (GPL), permitting the manufacture of Arduino boards and software distribution by anyone. Arduino boards are available commercially in preassembled form or as do-it-yourself (DIY) kits.Arduino board designs use a variety of microprocessors and controllers. It includes 14 digital I/O pins, six of which support PWM output, and six analog input pins offering 10-bit resolution. The Arduino uno is as shown in the fig.4.1. The board can be powered via a USB connection or an external power source, such as an adapter (7-12V) or battery, with onboard voltage regulators ensuring a stable 5V and 3.3V output. Communication capabilities include UART (serial), SPI, and I2C interfaces. A built-in USB to serial converter facilitates programming and serial communication through the USB-B port. The board also features a reset button, multiple power and ground pins, and LED indicators for power and digital pin 13. Additionally, the ICSP header allows for direct programming of the microcontroller. The inclusion of a 16 MHz crystal oscillator provides precise timing. These components collectively make the Arduino UNO a highly adaptable platform for a wide range of electronic projects.

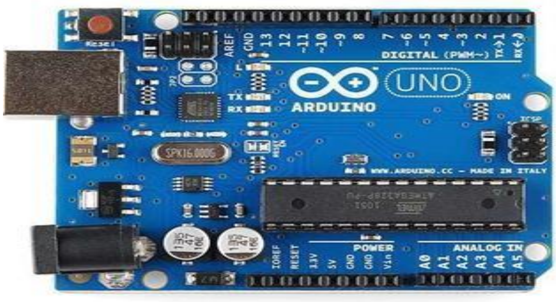

Fig.4.1: Arduino UNO

Arduino Uno specification:

Microcontroller	ATmega328P
Operating Voltage	5v
Input voltage	7-12v
Input voltage limit	6-20v
Digital I/O Pins	6
Analogue input Pins	6
DC current per I/O pins	20 mA
DC current for 3.3v Pin	50 mA
Flash Memory	Of which o.5KB is used
SRAM	2 KB
EEPROM	1KB
Clock Speed	16MHz
Length	68.6mm
Width	53.4nm
Weight	25g

Table.4.1: Arduino UNO specifications

The power pins are as follows:

- VIN: The input voltage to the Arduino board when it's using external power source (as opposed to 5 volts from the USB connection or other regulated power source). One can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- 5V: This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board.
- 3V3: A 3.3volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- GND: Ground pins.
- IOREF: This pin on the Arduino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin voltage and select the appropriate power source or enable voltage translators on the outputs to work with the 5V or 3.3V

Input &Output:

Each of the 14 digital pins on the Uno can be used as an input or output, using pin mode(), digital write (), and digital read () functions. They operate at 5 volts. Each pin can provide or receive 20 mA as recommended operating condition and has an internal pull- up resistor (disconnected by default) of 20-50k ohm. A maximum of 40 mA is the value that must not be exceeded on any I/O pin to avoid permanent damage to the microcontroller. The Arduino UNO pin specification as shown in fig.4.2.

In addition, some pins have specialized functions:

- Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the attach interrupt () function for details.

- PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the Analog write () function.
- SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication using the SPI library.
- LED: 13. There is a built-in LED driven by digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.
- TWI: A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library.

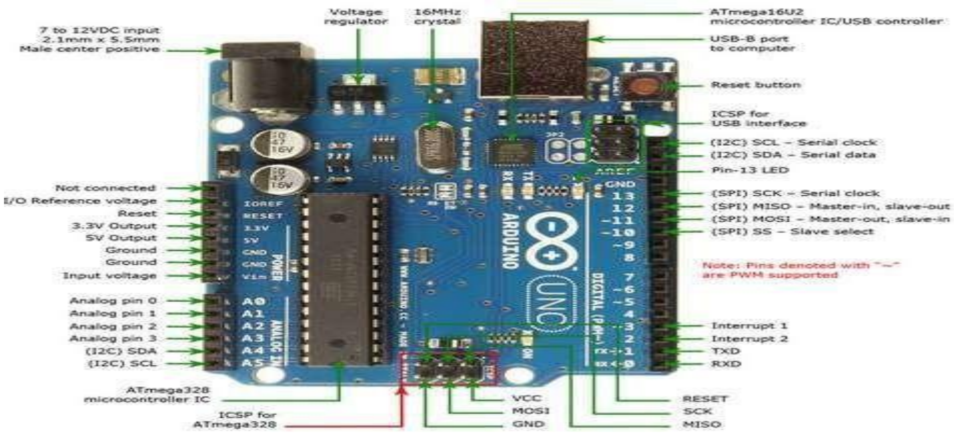

Fig.4.2: Arduino UNO pin specification

2) Power supply 12V

Fig.4.3: Power supply

The power supply as shown in fig.4.3 is an electronic device that provides a constant and stable output voltage regardless of variations in input voltage or load conditions It is designed to maintain a steady voltage level, ensuring that electronic components and systems operate reliably and within their specified parameters.

3) LCD Display

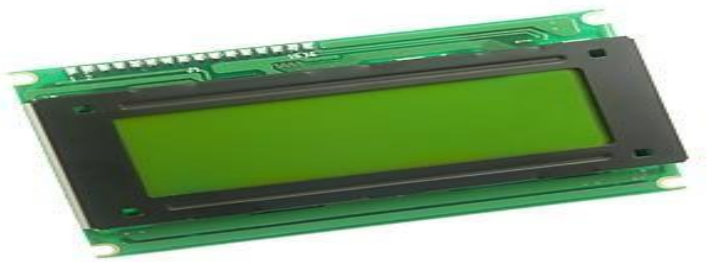

Fig.4.4: 16x2 LCD display

A **liquid-crystal display** (**LCD**) is as shown in fig4.4. flat-panel display or other electronically modulated optical device that uses the light-modulating properties of liquid crystals. Liquid crystals do not emit light directly, instead using a backlight or reflector to produce images in colour or monochrome. LCDs are available to display arbitrary images (as in a general-purpose computer display) or fixed images with low information content, which can be displayed or hidden, such as preset words, digits, and 7-segment displays, as in a digital clock. They use the same basic technology, except that arbitrary images are made up of a large number of small pixels, while other displays have larger elements. The LCD Pin Configuration as shown in Table 4.2 ,16*2 LCD demonstrates contains two lines likewise; there are 16 characters for each line. Each character is appeared by 5x7 pixel lattice. This LCD includes two registers, specifically, Order and Data. The charges select extras the charge bearings that are given to the LCD. A charge is a rule given to LCD to do a predefined errand like presenting it, clears its screen, sets the cursor position, controls show et cetera. The data enroll saves the data to be appeared on the LCD.

Terminal 1	GND
Terminal 2	+5V
Terminal 3	Mid terminal of potentiometer (for brightness control)
Terminal 4	Register Select (RS)

Terminal 5	Read/Write (RW)
Terminal 6	Enable (EN)
Terminal 7	DB0
Terminal 8	DB1
Terminal 9	DB2
Terminal 10	DB3
Terminal 11	DB4
Terminal 12	DB5
Terminal 13	DB6
Terminal 14	DB7
Terminal 15	+4.2-5V
Terminal 16	GND

Table.4.2: LCD Pin Configuration

4) Jumper Wires

Fig.4.5: Jumper wires

A **jumper wire** (also known as jumper, jumper wire, jumper cable, DuPont wire, or DuPont cable) is an electrical wire or group of them in a cable with a connector or pin at each end which is normally used to interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering. Individual jump wires are fitted by inserting their "end connectors" into the slots provided in a breadboard, the header connector of a circuit board, or a piece of test equipment.

5) DS18B20 Single wire Temperature sensor

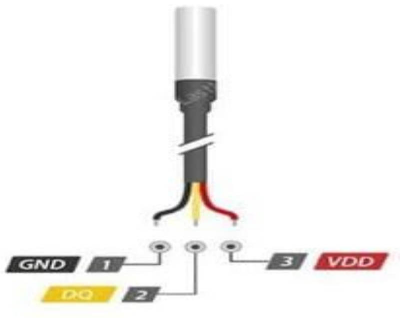

Fig.4.6: 1-wire DS18B20 temperature sensor

The 1-Wire DS18B20 Digital Temperature Sensor as shown in fig.4.6, is a widely used device for precise temperature measurement. This sensor communicates over a single data line, simplifying wiring and reducing the number of pins needed on microcontrollers. The DS18B20 offers a temperature range from -55°C to +125°C with an accuracy of ±0.5°C in the -10°C to +85°C range. It features unique 64-bit serial codes, allowing multiple sensors to be connected on the same 1-Wire bus. The sensor operates with a power supply range of 3.0V to 5.5V, making it versatile for various applications. Its ease of use, accuracy, and reliability make the DS18B20 a popular choice in projects like weather stations, thermostats, and process monitoring systems.

6) Relay

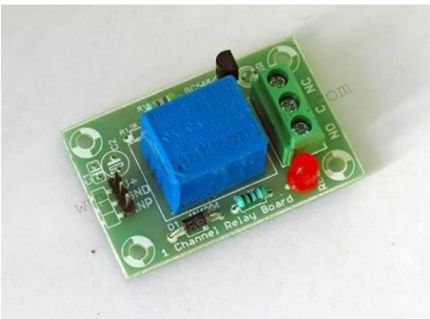

Fig.4.7: Relay

A relay is an electrically operated switch used to control a separate circuit with a 12V signal. It consists of a coil and a set of contacts When 12V is applied to the coil, it generates a magnetic field that activates the relay, causing the contacts to switch between positions. Relay is as shown in fig.4.7, Typically, there are normally open (NO) and normally closed (NC) contacts, with the common (COM) terminal serving as the connection point for the load. By applying 12V to the relay's coil, the NO contacts close, allowing current to flow through the load, while removing the voltage opens the contacts, cutting off the load. Properly wiring the relay and ensuring it can handle the voltage and current of the load is crucial, and using a fly back diode can protect the circuit from voltage spikes in inductive loads.

7) DC Fan

Fig.4.8: DC Fan

A 12V DC fan as shown in fig.4.8, is a cooling device commonly used in electronic and automotive applications to dissipate heat and maintain optimal operating temperatures for various components. Powered by a 12-volt direct current supply, these fans are efficient and versatile, making them ideal for use in computer systems, power supplies, and car interiors. They typically consist of a small motor that drives fan blades, creating airflow to cool down the surrounding environment. The compact size and low power consumption of 12V DC fans make them suitable for applications where space and energy efficiency are critical. Additionally, they are often equipped with features like variable speed control, allowing for adjustable airflow based on cooling needs.

CHAPTER 5

5.1 SOFTWARE REQUIREMENT AND CODE

ARDUINO IDE SOFTWARE:

The Arduino IDE software is based on the following theoretical concepts:

- Microcontroller Programming: Arduino boards are microcontrollers that can be programmed to perform specific tasks.
- C/C++ Programming Language: Arduino code is written in C/C++, which is a high-level programming language.
- Embedded Systems: Arduino boards are embedded systems, meaning they are designed to perform a specific task.
- Input/Output Operations: Arduino code reads input from sensors (e.g., temperature) and controls output devices (e.g., fans).
- Digital Signal Processing: Arduino code processes digital signals from sensors and controls digital outputs.
- Interrupts: Arduino code uses interrupts to handle events (e.g., temperature changes).
- Memory Management: Arduino code manages memory to optimize performance.
- Serial Communication: Arduino code uses serial communication to display data on the serial monitor.

CODE:

```
#include <LiquidCrystal.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#define ONE_WIRE_BUS 14
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
float temperature;
#include <SoftwareSerial.h>
int rs=13,en=12,d4=8,d5=9,d6=10,d7=11;
LiquidCrystal lcd(rs,en,d4,d5,d6,d7);
int fan=7;
void setup()
{
  pinMode(fan,OUTPUT);
  digitalWrite(fan,LOW);
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.clear();
  lcd.print("TEMPERATURE ");
  lcd.setCursor(0, 1);
  lcd.print("CONTROLLED FAN");
  delay(1000);
}
void loop()
{
    HUMIDITY_MEASUREMENT();
    }
 void HUMIDITY_MEASUREMENT()
{
     sensors.requestTemperatures();
      temperature = sensors.getTempCByIndex(0);
```

```
    Serial.print("$Temperature: ");
    Serial.print(temperature);
    Serial.println("#");
    lcd.clear();
  lcd.print("TEMP: ");
  lcd.setCursor(6, 0);
  lcd.print(temperature);
    delay(1000);
  if (temperature>30)
  {
   digitalWrite(fan,HIGH);
    lcd.clear();
  lcd.print("MORE TEMPERATURE ");
  lcd.setCursor(0, 1);
  lcd.print(" FAN ON");
  delay(1000);
// voltage();
   }
  else
  {
   digitalWrite(fan,LOW);
    lcd.clear();
  lcd.print("LESS TEMPERATURE ");
  lcd.setCursor(0, 1);
  lcd.print(" FAN OFF");
  delay(1000);
   }
  }
```

CHAPTER 6

6.1 RESULTS AND DISCUSSION

1. **Temperature Sensor:** The temperature sensor (DS18B20) converts the temperature into an electrical signal.
2. **Analog-to-Digital Converter (ADC):** The Arduino UNO's ADC converts the analog signal from the temperature sensor into a digital value.
3. **Microcontroller (MCU):** The Arduino UNO's MCU (ATmega328P) processes the digital temperature value and compares it to the set point.
4. **Digital Output:** If the temperature exceeds the set point, the MCU sends a digital HIGH signal (5V) to the relay module through a digital output pin.
5. **Relay Module:** The relay module receives the digital signal and switches its internal contacts to connect the fan to the power source (12V).
6. **Fan Speed Control:** The relay module can be configured to control the fan speed by Switching the fan on/off (simple on/off control)

CASE 1: When the temperature exceeds 30°C

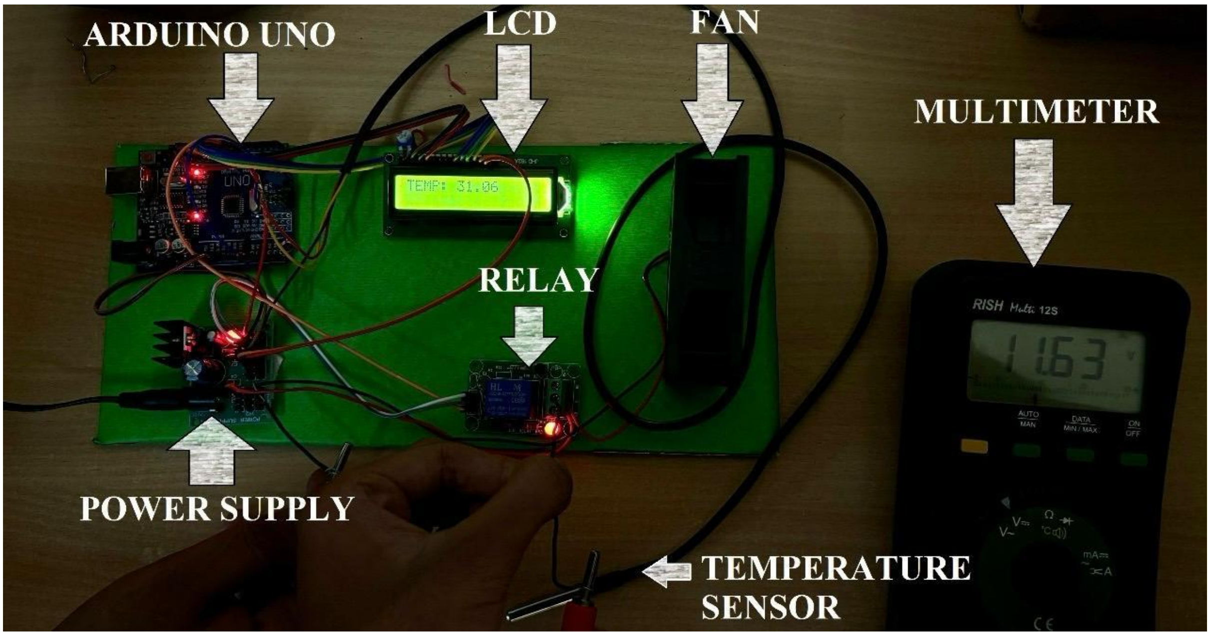

Fig.6.1: Temperature exceeds 30°C

When the temperature exceeds 30℃, a 12V supply is routed to the fan through relay, resulting in an output of 11.63V as indicated by the multimeter, as shown in the fig.6.1 above.

CASE 2: When the temperature is below 30℃

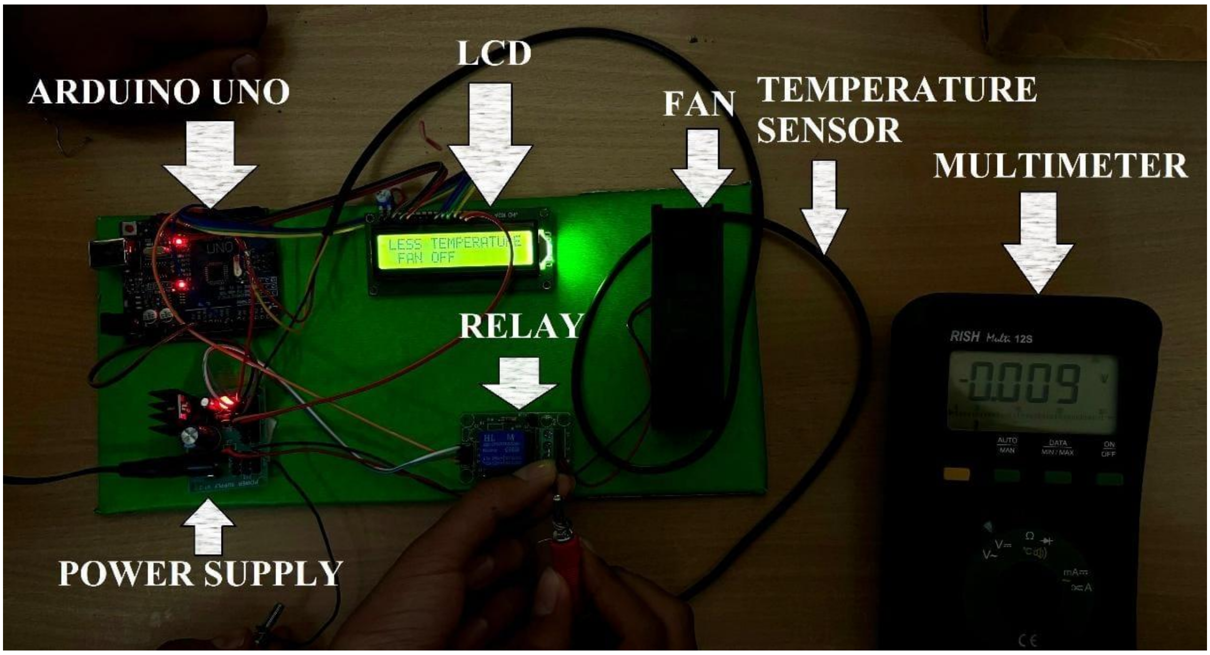

Fig.6.2: Temperature below 30℃

Similarly, when the temperature is below 30℃, output of 0.009V is indicated on the multimeter as shown in the above fig.6.2, where the fan will be turned off.

6.2 ADVANTAGES

- Energy Efficiency: The system optimizes fan speed to reduce energy consumption.
- Automated Fan Speed Control: Fan speed is adjusted automatically based on temperature changes.
- Improved Comfort: Maintains a comfortable environment by adjusting fan speed according to temperature.
- Real-time Temperature Monitoring: Displays real-time temperature data on the serial monitor.
- Cost-effective: Uses affordable Arduino components.
- Scalable: Can be scaled up or down depending on the application.
- Reliable: Durable and reliable for long-term use.
- Easy Installation: Simple to install and set up.
- User-friendly Interface: Easy-to-use interface for temperature monitoring.
- Data Analysis: Provides data for analysis to optimize fan performance and energy efficiency.

6.3 LIMITATIONS

- Limited Temperature Range: May not be suitable for extreme temperature ranges.
- Sensor Accuracy Limitations: Sensor accuracy may affect system performance.
- Fan Noise at High Speeds: Fan noise may be a issue at high speeds.
- System Complexity: System may be complex to understand and troubleshoot.
- Dependence on Sensor Calibration: Requires proper sensor calibration for accurate readings.
- Limited Fan Control: Only controls fan speed (on/off or speed).
- No Remote Monitoring/Control: No remote access to monitor or control the system.
- Limited User Customization: Limited options for user customization.
- Potential for Sensor Drift: Sensors may drift over time, affecting accuracy.
- Requires Maintenance: Requires regular maintenance to ensure optimal performance.

6.4 APPLICATIONS

- Home Automation: Suitable for home automation systems.
- Industrial Temperature Control: Can be used in industrial settings for temperature control.
- Greenhouse Climate Control: Ideal for greenhouse climate control.
- Server Room Cooling: Suitable for server room cooling systems.
- Data Center Cooling: Can be used in data center cooling systems.
- Automotive Temperature Control: Can be used in automotive temperature control systems.
- Aerospace Temperature Control: Suitable for aerospace temperature control applications.
- Medical Equipment Temperature Control: Can be used in medical equipment temperature control.
- Food Storage Temperature Control: Ideal for food storage temperature control.
- Laboratory Temperature Control: Suitable for laboratory temperature control applications.

CHAPTER 7

7.1 CONCLUSION

Basic idea of this project is to run the DC motor fan when temperature sensed by the temperature sensor is greater than threshold value. In this project we have used Arduino board for the programming of microcontroller through the USB. The microcontroller uses the hexadecimal file to execute the program. The temperature sensor output is connected to the microcontroller and it gives the output to the motor driver IC which runs the motor. The system effectively integrates Arduino with temperature sensing and motor control, adapting to changing temperature conditions with accuracy and reliability. With potential applications in HVAC, robotics, and industrial automation, this project showcases the innovative capabilities of Arduino-based systems in real-world applications.

7.2 FUTURE SCOPE

- Integration with IOT Devices: Potential for integration with IoT devices for remote monitoring/control.
- Remote Monitoring and Control: Future development of remote monitoring and control capabilities.
- Advanced Sensor Technologies: Potential for using advanced sensor technologies for improved accuracy.
- Machine Learning for Optimized Control: Future development of machine learning algorithms for optimized control.
- Multi-zone Temperature Control: Potential for controlling multiple zones with different temperature settings.
- Integration with HVAC Systems: Future development of integration with HVAC systems.
- Development of More Efficient Fan Designs: Potential for developing more efficient fan designs.
- Advanced User Interfaces: Future development of advanced user interfaces (e.g., mobile apps).
- Increased Scalability: Potential for increased scalability for large applications.

CHAPTER 8

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